VE401 Mid-2

Random Sample

sample range: $\max X_k - \min X_k$

Sample mean: $\bar{X} = \frac{1}{n}\sum X_k$ (1 more d.p)

Sample median: $\tilde{x}$

sample variance: (2 more d.p.)

$$S^2 = \frac{1}{n-1}\sum (X_k - \bar{X})^2$$

sample standard deviation S (1 more d.p)

Plots

Stem-and-Leaf Diagram

e.g.

Stem	Leaves
0	0011223345
1	0013445 68
2	1678
3	23

Stem units: 1000

*如果是double ~, 就把0~4和5~9的叶子分开

Histogram (柱状图)

① Find # of catagories

Data Set Size	# of Catagories
<16	insufficient
16~31	5
$2^{n-1} \sim 2^n - 1$	n

② Catagory length $= \frac{\text{range}}{n}$

③ Lower boundary $= \min X_k - \frac{1}{2}$ smallest decimal unit

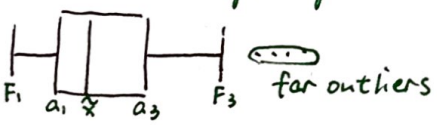
Boxplot

interquartile range: $iqr = q_3 - q_1$

inner fences $\begin{cases} f_1 = q_1 - \frac{3}{2} iqr \\ f_3 = q_3 + \frac{3}{2} iqr \end{cases}$

adjacent values $\begin{cases} a_1 = \min\{x_k \mid x_k \geq f_1\} \\ a_3 = \max\{x_k \mid x_k \leq f_3\} \end{cases}$

outer fences $\begin{cases} F_1 = q_1 - 3iqr \\ F_3 = q_3 + 3iqr \end{cases}$



Estimators (for θ: $\hat{\theta}$) $\begin{cases} \text{biased} \\ \text{unbiased} \end{cases}$

bias $= \theta - E[\hat{\theta}]$

$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = Var\hat{\theta} + (\text{bias})^2$

* $\bar{X}$ is an unbiased estimator for μ. $E[\bar{X}] = \mu$; $Var\,\bar{X} = \frac{\sigma^2}{n}$

* S^2 is an unbiased estimator for σ^2

* $M_k = \frac{1}{n}\sum X_i^k$ is an unbiased estimator for $E[X^k]$ ($VarX = E[X^2] - E[X]^2$)

Finding Estimators - Method of Moments

利用 M_k

Method of maximum likelihood

$L(\theta) = \prod f(x_i) \Rightarrow L(\hat{\theta}) = \max$

*取ln再求偏导

Distribution of Sample Mean

* $m_X(t) = m_Y(t)$ for all t in a $\varepsilon(0)$

$\Rightarrow f_X(x) = f_Y(x)$ for all x

* $Y = X_1 + X_2 \Rightarrow m_Y = m_{X_1} m_{X_2}$

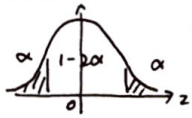
* $Y = \alpha + \beta X \Rightarrow m_Y = e^{\alpha T} m_X(\beta T)$

* $m_X(t) = E[e^{tX}]$. $E[X^k] = m_X^{(k)}(t)|_{t=0}$

$100(1-\alpha)\%$ confidence centered

$\begin{cases} \text{interval } [L_1, L_2] & P[L_1 \leq \theta \leq L_2] = 1-\alpha \\ \text{upper bound } L_U & P[\theta \leq L] = 1-\alpha \\ \text{lower bound } L_L & P[\theta \geq L] = 1-\alpha \end{cases}$

Def $\boxed{\alpha = P[Z \geq z_\alpha]}$



Interval Estimation for μ (normal)

$L_1, L_2 = \bar{X} \pm \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$;

$L_U = \bar{X} + \frac{z_\alpha \cdot \sigma}{\sqrt{n}}$; $L_L = \bar{X} - \frac{z_\alpha \cdot \sigma}{\sqrt{n}}$

Central Limit Theorem

* $X_1, \dots, X_n$ independent; $Y = \sum X$

$Z_n = \frac{Y - \sum \mu_j}{\sqrt{\sum \sigma_j^2}} \sim n(0,1)$ as $n \nearrow$

* $X_1, \dots, X_n$ same distribution

$Z_n = \frac{Y - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim n(0,1)$

Chi d. $X_n = \sqrt{\sum_{i=1}^n Z_i}$

$$f_{X_n}(y) = \begin{cases} \frac{2}{2^{n/2}\Gamma(\frac{n}{2})} y^{n-1} e^{-y^2/2}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Chi² d. $X_n^2 = \sum_{i=1}^n Z_i$

$$f_{X_n^2}(y) = \begin{cases} \frac{1}{2^{n/2}\Gamma(\frac{n}{2})} y^{n/2-1} e^{-y/2}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

* $X_\alpha^2 = \sum_{k=1}^n X_{Y_k}^2$ with $\alpha = \sum_{k=1}^n Y_k$

Theorem $X_1, \dots, X_n$ from $n(\mu, \sigma^2)$

① $\bar{X}$, S^2 independent ($\Leftrightarrow$)

② $\bar{X} \sim n(\mu, \sigma^2/n)$

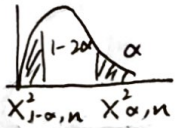
③ $\frac{(n-1)S^2}{\sigma^2} = \chi_{n-1}^2$

Interval Estimation for σ (normal)

$[L_1, L_2] = \left[\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}}\right]$;

$L_U = \frac{(n-1)S^2}{\chi^2_{1-\alpha, n-1}}$;

$L_L = \frac{(n-1)S^2}{\chi^2_{\alpha, n-1}}$.



The Student T-d. $T_Y = \frac{Z}{\sqrt{X_Y^2/Y}}$

$$f_{T_Y}(t) = \frac{\Gamma(\frac{Y+1}{2})}{\Gamma(\frac{Y}{2})\sqrt{\pi Y}} \left(1 + \frac{t^2}{Y}\right)^{-\frac{Y+1}{2}}$$

* $T_{n-1} = \frac{X - \mu}{S/\sqrt{n}}$

Interval Estimation for μ (normal)

$L_1, L_2 = \bar{X} \pm \frac{t_{\alpha/2, n-1} S}{\sqrt{n}}$;

$L_U = \bar{X} + \frac{t_{\alpha, n-1} S}{\sqrt{n}}$; $L_L = \bar{X} - \frac{t_{\alpha, n-1} S}{\sqrt{n}}$

Tolerance Limit / Interval

$(1-\alpha)100\%$ certain that $\delta \cdot 100\%$ of the population lies in $[L_1, L_2]$

if $X \sim n(\mu, \sigma)$.

$\bar{X} \pm k(n, \alpha, \delta) S$ one-sided ~

$\left.\begin{array}{l} (-\infty, \bar{X} + kS) \\ (\bar{X} - kS, \infty) \end{array}\right\}$ two-sided tolerance limit

else

$1 - \alpha = P[(X_{min}, X_{max}) \text{ covers } \delta 100\% \sim]$

$= 1 - n\delta^{n-1} + (n-1)\delta^n$

$n \approx \frac{1}{2} + \frac{1+\delta}{1-\delta} \cdot \frac{\chi^2_{1-\alpha, 4}}{4}$

Hypothesis Testing

$\left.\begin{array}{l} H_0: \theta \leq \theta_0 \quad H_1: \theta > \theta_0 \\ H_0: \theta \geq \theta_0 \quad H_1: \theta < \theta_0 \end{array}\right\}$ one-sided

$H_0: \theta = \theta_0 \quad H_1: \theta \neq \theta_0$ two-sided

$P[\text{reject } H_0 \mid H_0 \text{ is true}] = \alpha$ (level of significance)

$P[\text{reject } H_0 \mid H_0 \text{ is false}] = \text{power} = 1 - \beta$

$P[\text{accept } H_0 \mid H_0 \text{ is false}] = \beta$

$\begin{cases} \text{critical region} \\ \text{acceptance region} \end{cases}$

Relationship between $\delta, \alpha, \beta, n, \sigma$ (normal)

$\begin{cases} -z_\beta \approx z_{\alpha/2} - \delta\sqrt{n}/\sigma \\ n \approx \frac{(z_{\alpha/2} + z_\beta)^2 \sigma^2}{\delta^2} \end{cases}$

used in significance test

P-value

$P[\text{observing sth. at least as extreme as the null value}]$

if P-value is small, reject H_0

Acceptance Sampling

$H_0: \pi \leq \pi_0 \quad H_1: \pi > \pi_0$ proportion of defects

$\begin{cases} \alpha \text{ producer's risk} = \sum_{d \geq c} \frac{\binom{n}{d}\binom{N-n}{n-d}}{\binom{N}{n}} \\ \beta \text{ consumer's risk} = \sum_{d \geq c} \frac{\binom{n}{d}\binom{N-n}{n-d}}{\binom{N}{n}} \end{cases}$

c acceptance number

Test	Z-Test	T-Test	Chi-Squared-Test
H_0:	$\mu=\mu_0$	$\mu=\mu_0$	$\sigma=\sigma_0$
Statistic	$Z=\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$	$T_{n-1}=\frac{\bar{X}-\mu_0}{S/\sqrt{n}}$	$\chi^2_{n-1}=\frac{(n-1)S^2}{\sigma_0^2}$
In favor of H_1:	$\mu\neq\mu_0$ if $\lvert Z\rvert>z_{\alpha/2}$	$\mu\neq\mu_0$ if $\lvert T_{n-1}\rvert>t_{\alpha/2,n-1}$	$\sigma\neq\sigma_0$ if $\chi^2_{n-1}>\chi^2_{\alpha/2,n-1}$ or $<\chi^2_{1-\alpha/2,n-1}$
	$\mu>\mu_0$ if $Z>z_\alpha$	$\mu>\mu_0$ if $T_{n-1}>t_{\alpha,n-1}$	$\sigma>\sigma_0$ if $\chi^2_{n-1}>\chi^2_{\alpha,n-1}$
	$\mu<\mu_0$ if $Z<-z_\alpha$	$\mu<\mu_0$ if $T_{n-1}<-t_{\alpha,n-1}$	$\sigma<\sigma_0$ if $\chi^2_{n-1}<\chi^2_{1-\alpha,n-1}$
(纵坐标为β) OC Curve	$d=\frac{\lvert\mu-\mu_0\rvert}{\sigma}$	$d=\frac{\lvert\mu-\mu_0\rvert}{s}$	$\lambda=\frac{\sigma}{\sigma_0}$

Sign Test for Median

$Q_+=\#\{X_k: X_k>M_0\}$ $Q_-=\#\{X_k: X_k<M_0\}$

we reject $H_0: M=M_0$ in favor of

$H_1: M>M_0$ if Q is small

$H_1: M<M_0$ if Q_+ is small

$H_1: M\neq M_0$ if $\min\{Q_-,Q_+\}$ is small

Wilcoxon Signed Rank Test

$W_+=\sum_{R_i>0}R_i$ $\lvert W_-\rvert=\sum_{R_i<0}\lvert R_i\rvert$

we reject $H_0: M=M_0$ in favor of

$H_1: M>M_0$ if W_- is small

$H_1: M<M_0$ if W_+ is small

$H_1: M\neq M_0$ if $\min\{W_-,W_+\}$ is small

Wilcoxon Rank-Sum Test

$f_X(x)=f_Y(x+\delta)$

$W_X=\sum R_X$ $W_Y=\sum R_Y$

we reject $H_0: M_X=M_Y$ in favor of

$H_1: M_X\neq M_Y$ ... W_X normal

$H_1: M_X\gtrless M_Y$ ... $\mu=\frac{n_1(n_1+n_2+1)}{2}$ $\sigma^2=\frac{n_1n_2(n_1+n_2+1)}{12}$

Test for proportions

$H_0: p=p_0$

$Z=\frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)}/\sqrt{n}}$

$H_1: p\neq p_0$ if $\lvert Z\rvert>z_{\alpha/2}$

$p>p_0$ if $Z>z_\alpha$

$p<p_0$ if $Z<-z_\alpha$

Test for comparing two proportions

$H_0: p_1-p_2=(p_1-p_2)_0$

$Z=\frac{\hat{p}_1-\hat{p}_2-(p_1-p_2)_0}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n}+\frac{\hat{p}_2(1-\hat{p}_2)}{n}}}$

$H_1: p_1-p_2\neq(p_1-p_2)_0$

$>$

$<$

Test for equality of p

$H_0: p_1=p_2$

$Z=\frac{\hat{p}_1-\hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1}+\frac{1}{n_2})}}$

$H_1: p_1-p_2\neq 0$

$>$

$<$

$\hat{p}=\frac{n_1\hat{p}_1+n_2\hat{p}_2}{n_1+n_2}$

Comparing 2 μ.

(known σ)

$H_0: \mu_1=\mu_2$.

$Z=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}}$

... ...

$d=\frac{\lvert\mu_1-\mu_2\rvert}{\sqrt{\sigma_1^2+\sigma_2^2}}$

$n=\frac{\sigma_1^2+\sigma_2^2}{\sigma_1^2/n_1+\sigma_2^2/n_2}$

Pooled T-Test

(unknown equal σ)

$S_p^2=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$

$\chi^2_{n_1+n_2-2}=\frac{(n_1-1)S_1^2}{\sigma_1}+\frac{(n_2-1)S_2^2}{\sigma_2}$

$T_{n_1+n_2-2}=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{S_p^2(1/n_1+1/n_2)}}$

$d=\frac{\lvert\mu_1-\mu_2\rvert}{2\sigma}$ *先验证

$n^*=2n-1$ equal σ

(unknown unequal)

$H_0: \mu_1-\mu_2=(\mu_1-\mu_2)_0$

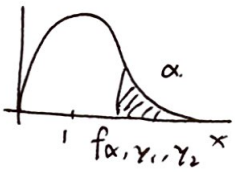
$T_\gamma=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)_0}{\sqrt{S_1^2/n+S_2^2/n}}$

$\gamma=\frac{(S_1^2/n_1+S_2^2/n_2)^2}{\frac{(S_1^2/n_1)^2}{n_1-1}+\frac{(S_2^2/n_2)^2}{n_2-1}}$

F-distribution

$F_{\gamma_1,\gamma_2}=\frac{\chi^2_{\gamma_1}/\gamma_1}{\chi^2_{\gamma_2}/\gamma_2}$

$f_{\gamma_1,\gamma_2}(x)=\gamma_1^{\gamma_1/2}\gamma_2^{\gamma_2/2}\frac{\Gamma(\frac{\gamma_1+\gamma_2}{2})}{\Gamma(\frac{\gamma_1}{2})\Gamma(\frac{\gamma_2}{2})}\frac{x^{\gamma_1/2-1}}{(\gamma_1x+\gamma_2)^{(\gamma_1+\gamma_2)/2}}$



$P[F_{\gamma_1,\gamma_2}>f_{\alpha\gamma_1\gamma_2}]=\alpha$.

$f_{1-\alpha,\gamma_1,\gamma_2}=\frac{1}{f_{\alpha\gamma_1\gamma_2}}$

* If $\sigma_1^2=\sigma_2^2$, $F_{n_1-1,n_2-1}=\frac{\sigma_2^2S_1^2}{\sigma_1^2S_2^2}$

	Geo	Bio	Pas	Hyp	Poi
f_X	$q^{x-1}p$	$\binom{n}{x}p^xq^{n-x}$	$\binom{x-1}{r-1}p^rq^{x-r}$	$\frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}$	$\frac{k^xe^{-k}}{x!}$
E	$\frac{1}{p}$	np	$\frac{r}{q}$	$\frac{nr}{N}$	k
Var	$\frac{q}{p^2}$	npq	$\frac{rq}{p^2}$	$\frac{nr}{N}\frac{N-r}{N}\frac{N-n}{N-1}$	k

($k=\lambda t$)

exponential d.

$f_X(x)=\frac{1}{\beta}e^{-x/\beta}$ $(x>0)$

$E[X]=\beta$ $Var X=\beta^2$

proportions

$X=\begin{cases}1, & \text{trait}\\ 0, & \text{no trait}\end{cases}$

$p=\frac{1}{N}\sum x$ $\hat{p}=\frac{1}{n}\sum x$

$\hat{p}\sim n(p,p(1-p)/n)$ (approximately)

$100(1-\alpha)\%$ confidence interval for p

$\hat{p}\pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$

Given $d=z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$

$n=\frac{z^2_{\alpha/2}\hat{p}(1-\hat{p})}{d^2}\leq\frac{z^2_{\alpha/2}}{4d^2}$

For larger n, $\hat{p}_1-\hat{p}_2\sim n(p_1-p_2,\frac{p_1(1-p_1)}{n_1}+\frac{p_2(1-p_2)}{n_2})$

$100(1-\alpha)\%$ confidence interval of (p_1-p_2)

$\hat{p}_1-\hat{p}_2\pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

F-Test

$H_0: \sigma_1=\sigma_2$

$F_{n_1-1,n_2-1}=\frac{S_1^2}{S_2^2}$

$H_1: \sigma_1=\sigma_2$ $\frac{S_1^2}{S_2^2}>f_{\alpha/2,n_1-1,n_2-1}$ or $\frac{S_2^2}{S_1^2}<f_{\alpha/2,n_2-1,n_1-1}$

$\sigma_1>\sigma_2$ $\frac{S_1^2}{S_2^2}>f_{\alpha,n_1-1,n_2-1}$

$\sigma_1<\sigma_2$ $\frac{S_2^2}{S_1^2}>f_{\alpha,n_1-1,n_2-1}$

$\lambda=\frac{\sigma_1}{\sigma_2}$

Calculator CASIO 991ES

MODE 3 1 ← Data.

AC Shift 1

2015.7